

Network Intrusion Detection on the CICIoT2023 Dataset
Using Random Forest Classification

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Abstract:

This project looked at using machine learning for spotting intrusions in IoT traffic. They used the CICIoT2023 set which came from a testbed with 105 devices and attacks from other hacked ones. A Random Forest model was run on an 8-class version that included benign traffic plus seven attack types. It was meant as a direct comparison to an earlier project on CICIDS2017 with the exact same setup so differences could be blamed on the data change. Overall, the scores looked really high but when broken down by class the model struggled badly with Brute Force and Web attacks. I think that came mostly from how few samples there were for those rather than anything else going wrong. The features it leaned on were different too. This time it mostly used one timing thing called Inter Arrival Time to sort the attacks instead of packet length stats like the older model did. That stood out when they checked the importance rankings. Some parts of the results matched what the original dataset paper said and other studies found as well. It seems like the imbalance issue is pretty common in this kind of data. Maybe more samples for the rare attacks would help but the project did not test that.

Objectives:

- Train a machine learning model capable of classifying IoT network traffic into eight categories: Benign, DDoS, DoS, Mirai, Recon, Spoofing, Web, and Brute Force.
- Directly compare methodology and results against an earlier CICIDS2017 project, holding the model and pipeline constant so the dataset is the only varying factor.
- Critically evaluate model performance beyond surface-level accuracy, using precision, recall, F1-score, and confusion matrices, with particular attention to macro-averaged metrics given severe class imbalance.
- Investigate the underlying causes of unusually high aggregate performance scores rather than accepting them at face value.
- Identify which network traffic features most influence the model's predictions, and assess whether these features reflect genuine IoT attack behaviour or dataset-specific artifacts.
- Cross-validate findings against the original dataset paper and independent published literature.

Dataset:

This project is working with CICIoT2023 which was created by the Canadian Institute for Cybersecurity. They built it using a real testbed of 105 IoT devices including

cameras speakers sensors and some Raspberry Pi nodes. These devices were set up to act as both attackers and targets. It seems different from something like CICIDS2017 since the attacks here come from compromised IoT devices going after other IoT devices rather than external actors. That lines up better with current botnet activity such as Mirai.

The original release is huge with over 46 million records split across many CSV files. This project instead uses a pre processed single file version called the derived CICIoT 2023 datasets from El Balbali and Abou El Kalam in 2025. It takes the 33 attack types and puts them into 8 broader groups using the 8class data csv file. That file has 3317513 rows and 47 columns with 46 numeric features plus the label. It feels like the right size for this kind of work.

The label column in this derived file contains integers (0–7) which map respectively to Benign, BruteForce, DDoS, DoS, Mirai, Recon, Spoofing, and Web

Procedure:

1. Data Loading and Exploration:

The dataset was loaded using pandas, and column names were stripped of extraneous whitespace. The label column was examined via `value_counts()` to confirm there were exactly 8 distinct classes, and a `label_name` column was added by mapping the integer codes to text names (see mapping note above) for readability in plots and reports.

2. Data Cleaning:

Infinite values were replaced with NaN, and rows containing any NaN were dropped. No rows were removed, the dataset contained zero missing or infinite values prior to cleaning, and the class counts before and after this step were identical, confirming no class was disproportionately affected.

3. Data Preparation:

Features (x) were separated from the target label (y). Both the numeric label column and the added `label_name` text column were excluded from x to prevent label leakage. All 46 remaining feature columns were confirmed numeric (`float64`), so no categorical encoding was required.

The dataset was split into training (70%) and test (30%) sets using stratified sampling, preserving class ratios in both subsets, particularly important given that the rarest class (Brute Force) contains only 900 rows total.

4. Model Training:

A Random Forest classifier (100 estimators, random_state=42) was used, the same algorithm and hyperparameters as the earlier CICIDS2017 project. This was a deliberate methodological choice: keeping the model constant isolates the dataset as the single variable under comparison, so any differences in behaviour can be attributed to the change in traffic environment rather than a change in modelling approach.

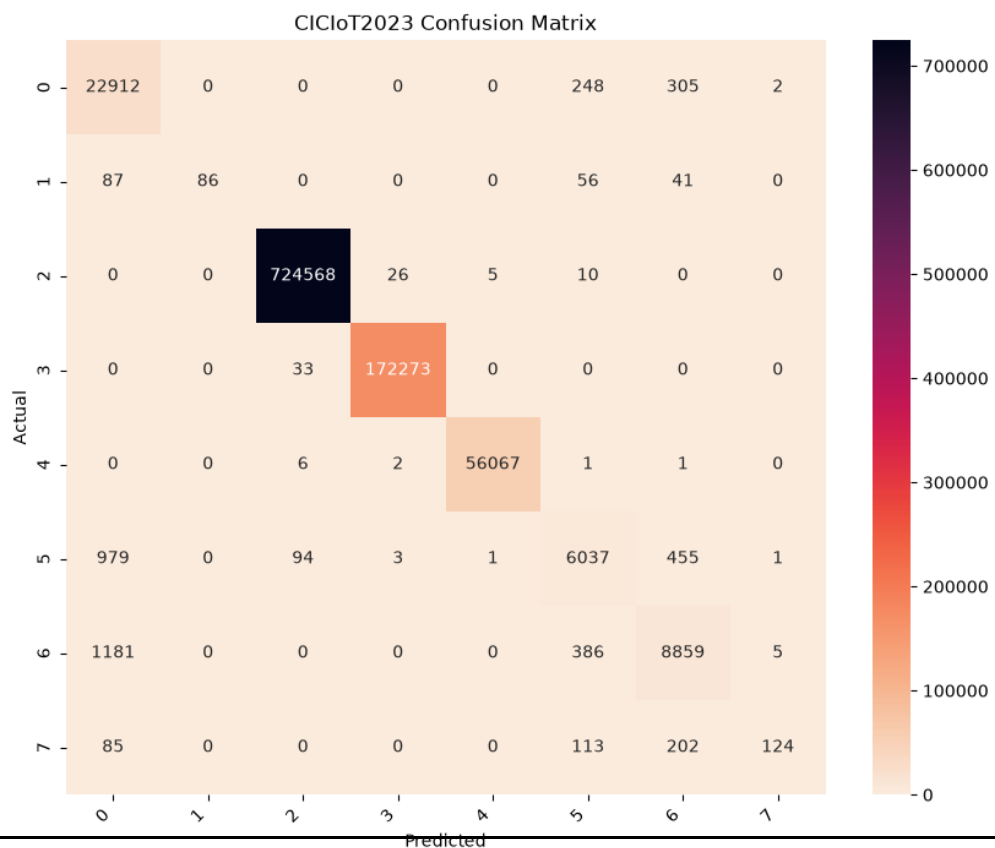
5. Evaluation:

Initial results:

Initial evaluation produced near-perfect aggregate metrics:

	precision	recall	f1-score	support
Benign	0.91	0.98	0.94	23,467
BruteForce	1.00	0.32	0.48	270
DDoS	1.00	1.00	1.00	724,609
DoS	1.00	1.00	1.00	172,306
Mirai	1.00	1.00	1.00	56,077
Recon	0.88	0.80	0.84	7,570
Spoofing	0.90	0.85	0.87	10,431
Web	0.94	0.24	0.38	524
accuracy			1.00	995,254
macro avg	0.95	0.77	0.81	995,254
weighted avg	1.00	1.00	1.00	995,254

Confusion Matrix:



Given how unusual near-perfect scores are in applied machine learning, and consistent with the skepticism applied in the earlier CICIDS2017 project, this result was treated as a finding to investigate further.

Investigating the near-perfect scores:

An exact-duplicate check (based on feature columns) found that 15.3% of all rows (506,535 of 3,317,513), and 18.9% of DDoS rows specifically, were exact duplicates of other rows, raising concern that a plain random train/test split could place near-identical flows on both sides of the split, artificially inflating scores through memorization rather than genuine generalization. This concern is directly supported by prior literature documenting this exact leakage mechanism in CICIoT2023, attributed to automated attack-generation scripts producing highly repetitive flow records.

To test this hypothesis directly, duplicate rows were removed and the model was retrained from scratch on the deduplicated data (2,810,978 rows), using an identical stratified split and identical model settings.

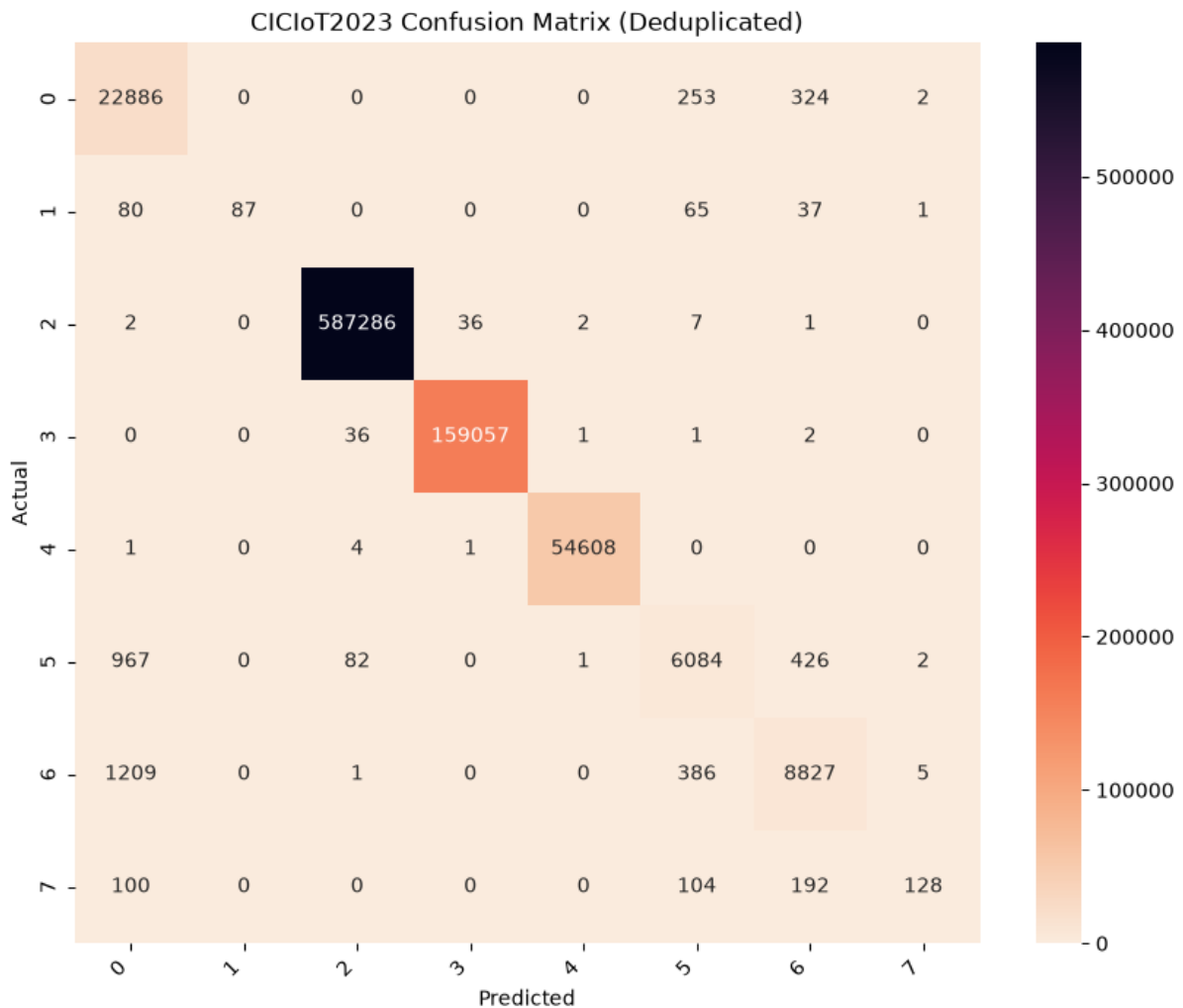
Results after de-duplication:

	precision	recall	f1-score	support
Benign	0.91	0.98	0.94	23,465
BruteForce	1.00	0.32	0.49	270
DDoS	1.00	1.00	1.00	587,334
DoS	1.00	1.00	1.00	159,097
Mirai	1.00	1.00	1.00	54,614
Recon	0.88	0.80	0.84	7,562
Spoofing	0.90	0.85	0.87	10,428
Web	0.93	0.24	0.39	524
accuracy			0.99	843,294
macro avg	0.95	0.77	0.82	843,294
weighted avg	0.99	0.99	0.99	843,294

Performance was virtually unchanged after de-duplication. This is an important negative result, it indicates that DDoS, DoS, and Mirai's near-perfect scores reflect genuine statistical separability, likely because flood-style attacks push features like packet rate and flag counts to extreme, saturating values that simply don't overlap with normal or other attack traffic, rather than memorization of duplicate records. Notably, BruteForce and Web contained zero duplicate rows to begin with, so their weak performance (discussed below) cannot be a duplication artifact either; it points to a separate cause.

Confusion Matrix:

[P.T.O]



Class-level weaknesses:

Two classes stood out as clear weak spots:

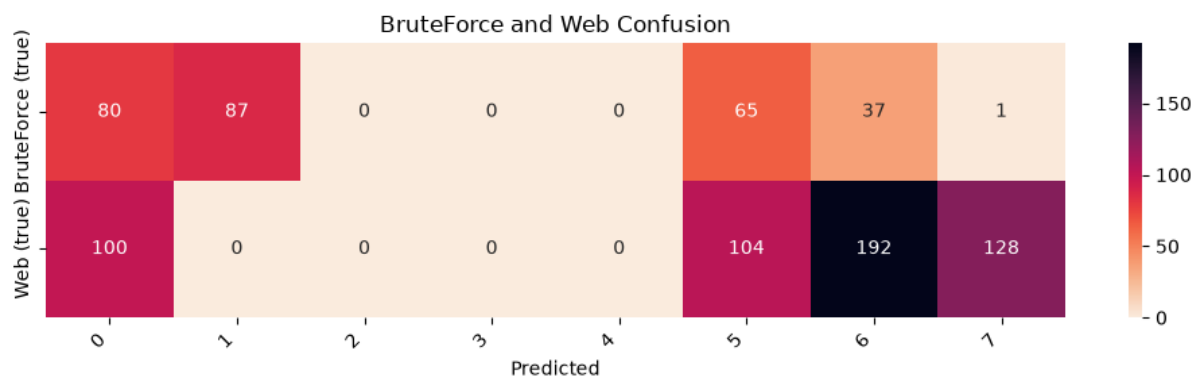
- **Brute Force**: recall of 0.32 (only 87 of 270 test samples correctly identified)
- **Web**: recall of 0.24 (only 128 of 524 test samples correctly identified)

Unlike the CICIDS2017 project, where the weak class (Bots) was confused almost exclusively with Normal traffic, this dataset shows a more distributed confusion pattern:

- **Brute Force** was most often confused with **Benign**, secondarily with Recon and Spoofing.
- **Web** was most often confused with **Spoofing**, secondarily with Recon and Benign.
- **Recon** and **Spoofing** frequently confused with each other, in addition to both leaking into Benign.

This suggests that several of these IoT attack categories (Recon, Spoofing, Web, Brute Force) share overlapping low-and-slow traffic signatures with each other, not just with normal traffic, a more complex confusion structure than was observed in CICIDS2017.

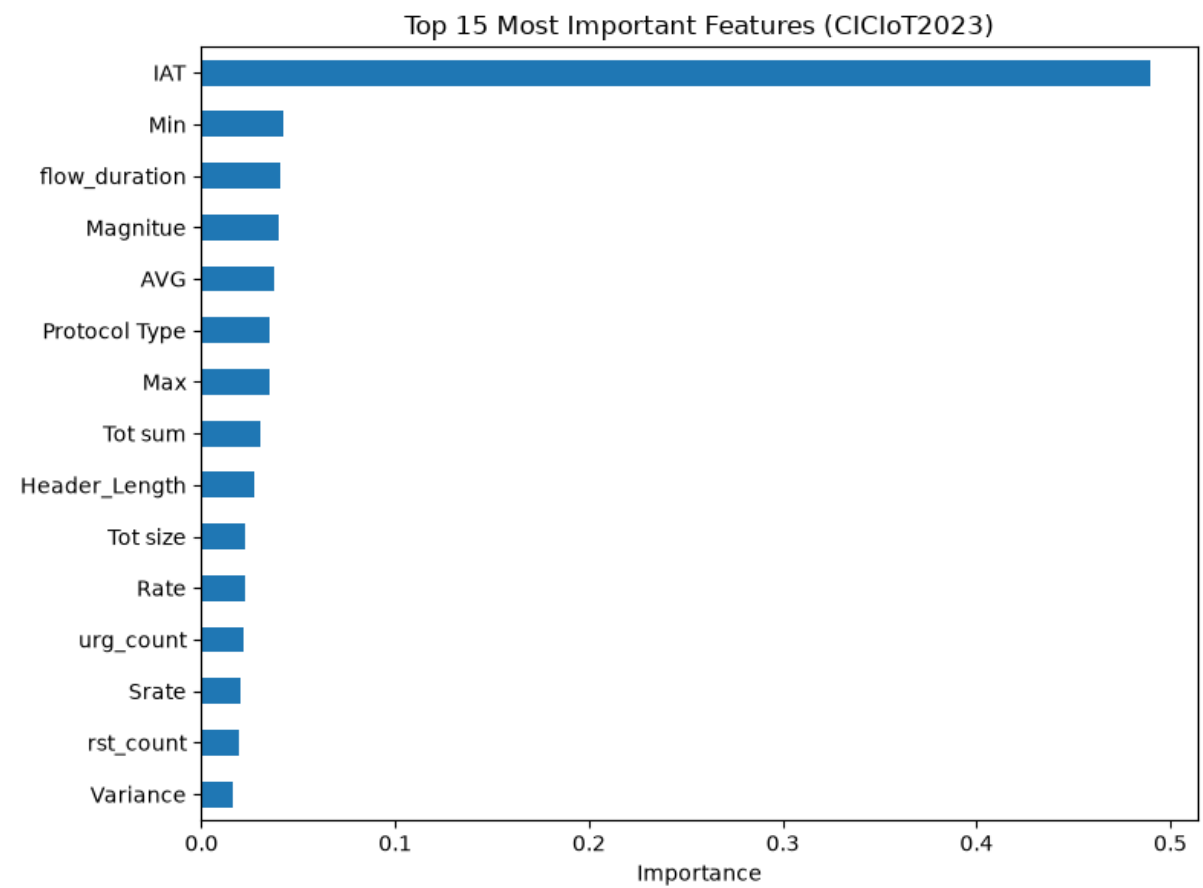
Given that Brute Force and Web are also the two smallest classes by a wide margin (900 and 1,747 total rows respectively, versus hundreds of thousands or millions for the other classes), and given that de-duplication ruled out data leakage as an explanation, **this weakness is best attributed to severe class scarcity**, the model simply does not see enough examples of these attack types to learn a reliable decision boundary, and there are too few test examples to score that boundary precisely either.



6. Feature Importance Analysis:

Feature importances were extracted from the deduplicated Random Forest model.

Top 15 features:



This is the central comparison point against the CICIDS2017 project. In that project, packet-length statistics (Bwd Packet Length Std, Packet Length Variance, etc.) collectively accounted for roughly a third of total feature importance, and no single feature dominated.

Here, the result is markedly different, a single feature i.e. Inter-Arrival Time (IAT) accounts for 0.49, roughly ten times the importance of the next-ranked feature, and dwarfs the combined packet-length cluster (Min, AVG, Max, Tot sum, Header_Length, Tot size, Variance), which together total only about 0.21.

This makes behavioural sense: automated flood attacks (DDoS, DoS, Mirai, which together make up over 95% of this dataset) fire packets in tight, scripted, highly regular loops, producing very small and very consistent inter-arrival times. Benign IoT traffic, driven by real device and human-triggered behaviour, has much more naturally irregular timing. This ties directly to the earlier finding that DDoS/DoS/Mirai separability survived de-duplication, timing regularity, not packet size, appears to be the primary signal the model is exploiting.

7. Cross-Validation Against External Literature:

To assess whether these findings were an artifact of this specific pipeline, results were compared against the original CICIoT2023 dataset paper (Neto et al., 2023) and subsequent independent studies:

- **IAT dominance** is independently reported in multiple studies of this dataset, including work noting that Inter-Arrival Time provides most of the discriminative information for tree-based models on CICIoT2023.
- **The Brute Force / Web confusion pattern closely replicates the original dataset paper's own Random Forest results**, which report that web-based attacks are usually classified as benign, Recon, or Spoofing, with the same pattern occurring for Brute Force, an almost exact match to the confusion pattern observed in this project.
- **Un-resampled, imbalance-sensitive performance on rare classes** is consistent with broader literature reporting substantially lower macro-F1 scores on 8-class CICIoT2023 without resampling techniques such as SMOTE, compared to near-perfect binary classification scores.

This convergence with independent, previously published results strengthens confidence that the patterns observed here reflect genuine properties of IoT attack traffic and Random Forest behaviour on this dataset, rather than errors specific to this pipeline.

Key Findings:

- The model achieves near-perfect aggregate performance, but, as with CICIDS2017, this is substantially driven by three dominant classes (DDoS, DoS, Mirai) that together make up over 95% of the data.
- A duplicate-record leakage hypothesis was directly tested and **ruled out**: removing ~15% duplicate rows and retraining produced virtually no change in results, indicating that majority-class separability is genuine rather than a memorization artifact.
- Two classes i.e. Brute Force (0.32 recall) and Web (0.24 recall) show clear, meaningful weaknesses. Unlike CICIDS2017's single weak class, these two contain **zero duplicate rows**, ruling out leakage as their cause and pointing instead to **severe class scarcity** (900 and 1,747 total samples).
- Confusion patterns for the weak classes are more distributed than in CICIDS2017, spreading across Benign, Recon, and Spoofing rather than concentrating almost entirely in one "normal traffic" bucket, suggesting genuine behavioural overlap between several IoT attack types.
- Feature importance reveals a fundamentally different reliance pattern than CICIDS2017, a single timing feature (IAT) dominates here, rather than a cluster of packet-length statistics. This is directly explainable by the automated, tightly-scripted nature of flood-style IoT attacks.
- These findings closely replicate results independently reported in the original CICIoT2023 paper and subsequent literature, providing external validation beyond this project's own pipeline.

Limitations and Future Work:

- **Random train/test split**: this evaluation uses a random (stratified) split, which may allow flows from the same attack session to appear in both training and test sets even after exact-duplicate removal (near-duplicate, non-identical rows were not addressed). A stronger evaluation would use a temporal or session-based split.
- **Severe class scarcity**: Brute Force and Web's poor recall reflects a fundamental data availability problem rather than a fixable modelling error. Future work could explore SMOTE, other oversampling techniques, or class-weighted training, which prior literature reports substantially improves minority-class detection on this dataset.

- **Automated, lab-generated traffic:** CICIoT2023's attacks were generated by scripted, automated tools in a controlled testbed, which may produce more statistically consistent signatures than real-world, more varied attack traffic.

References:

- Neto, E. C. P., Dadkhah, S., Ferreira, R., Zohourian, A., Lu, R., & Ghorbani, A. A. (2023). CICIoT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment. *Sensors*, 23(13), 5941.
- El Balbali, H., & Abou El Kalam, A. (2025). Derived CICIoT 2023 datasets. *Mendeley Data*, V1. DOI: 10.17632/x36s52ybfw.1.
- Sharafaldin, I., Lashkari, A. H., & Ghorbani, A. A. (2018). Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization. *4th International Conference on Information Systems Security and Privacy (ICISSP)*, Portugal, 108–116.